

Technical Project Overview

LeGrad-Inspired Temporal Explanations for Speech Emotion Recognition

gradient-based-speech_xai

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1 Project Overview

This project compares two approaches for explaining a pretrained speech emotion recognition classifier. SpeechXAI provides human-readable explanations based on word-aligned audio segments. The proposed LeGrad-inspired method will derive temporal relevance from gradients associated with the attention mechanisms of a wav2vec 2.0 encoder. The classification task uses speech samples and emotion annotations from IEMOCAP.

The quantitative evaluation uses deletion by silence masking. Important audio regions are replaced by silence while the original signal length is preserved, and the resulting confidence drop for the classifier’s original predicted class is measured. The central methodological decision is to compare word-level SpeechXAI explanations with time-bin or token-level LeGrad explanations under the same total occluded duration.

2 Sources and External References

- **Dataset: IEMOCAP.** The local data follows the five-session IEMOCAP release from the USC Signal Analysis and Interpretation Laboratory. The corpus contains speech, transcripts, and categorical emotion annotations. Project source: [USC IEMOCAP website](#). Dataset paper: [Busso et al., “IEMOCAP: Interactive Emotional Dyadic Motion Capture”](#).
- **Pretrained classifier.** The project uses [superb/wav2vec2-base-superb-er](#), the Hugging Face wav2vec 2.0 emotion-recognition checkpoint loaded by the SpeechXAI repository for IEMOCAP. The model exposes four output classes: anger, happiness/excitement, neutral, and sadness.
- **wav2vec 2.0.** The encoder architecture originates from Baevski et al., [“wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations”](#).
- **SUPERB.** The active checkpoint follows the SUPERB speech-processing benchmark described by Yang et al., [“SUPERB: Speech Processing Universal PERFORMANCE Benchmark”](#).
- **SpeechXAI.** The word-level explanation method follows Pastor et al., [“Explaining Speech Classification Models via Word-Level Audio Segments and Paralinguistic Features,” EACL 2024](#). Repository: [elianap/SpeechXAI](#).
- **WhisperX alignment.** SpeechXAI obtains word timestamps with Bain et al., [“WhisperX: Time-Accurate Speech Transcription of Long-Form Audio,” Interspeech 2023](#). WhisperX combines VAD-guided segmentation, Whisper transcription, and external phoneme alignment.

- **LeGrad.** The proposed temporal method is inspired by Boussselham et al., “[LeGrad: An Explainability Method for Vision Transformers via Feature Formation Sensitivity](#)”. Repository: [WalBouss/LeGrad](#).

The current project configuration uses IEMOCAP. No separate YAMLCap dataset source is configured.

3 Notebook-by-Notebook Technical Overview

3.1 Notebook 01 — SpeechXAI Single Example

Purpose

Establish a complete SpeechXAI explanation and deletion-evaluation workflow on one IEMOCAP utterance before integrating the LeGrad-inspired method.

Technical idea

SpeechXAI uses WhisperX to transcribe the utterance and align each word with its actual start and end time. The same timestamp objects are passed into SpeechXAI’s leave-one-out explainer, ensuring that each attribution score refers to the word interval shown in the visualization. SpeechXAI measures importance as the confidence change caused by replacing a word region with silence. The paper describes zeroing the samples inside the WhisperX word interval. The released repository silences a larger interval: it starts 100 ms before the WhisperX word start and ends 40 ms after the WhisperX word end. Thus, the attribution assigned to one word is calculated after silencing the word together with these adjacent audio samples.

Decisions made

- A pretrained wav2vec 2.0 emotion classifier is used so the project can focus on explanation quality rather than classifier training.
- One utterance is used to keep this notebook focused on the explanation workflow and visualization.
- The original predicted class is the attribution and deletion target.
- Silence replacement preserves the waveform length.
- SpeechXAI’s WhisperX transcription is the only source of word text and timing; no character-based or uniform timing approximation is used.
- Transcription is performed once, and the returned word segments are reused by the explainer and plot so words, timestamps, and scores remain index-aligned.
- WhisperX’s per-word transcription confidence is retained in the attribution table so uncertain alignments remain visible during interpretation.
- The paper’s Section 2.1 and Appendix B.1 describe zeroing word segments derived from WhisperX timestamps to preserve the original recording length. They do not specify temporal padding around those segments.

- The released repository’s `utils_removal.py` defines `a=100` and `b=40`. For a WhisperX word interval $[t_{\text{start}}, t_{\text{end}}]$, `remove_word` and `remove_specified_words` silence $[t_{\text{start}} - 100 \text{ ms}, t_{\text{end}} + 40 \text{ ms}]$.
- The notebook reproduces the repository implementation rather than silently replacing it with the paper-level interval description.
- The visualization distinguishes two regions: the lexical interval reported by WhisperX and the larger interval actually silenced by the repository.
- A separate overlap diagnostic displays every effective mask on its own row and computes sample-level mask depth. Regions covered by two or more masks are highlighted explicitly.
- Attribution bars use SpeechXAI’s raw confidence-drop values. They are not rescaled to make the largest word equal to one, because that would hide the magnitude of the model response.
- The waveform-loading path used by the project is checked against SpeechXAI’s pydub-loading path. Both must produce the same predicted class and probability vector before the explanation is accepted.
- Word intervals must be chronological, have positive duration, remain inside the audio, and correspond one-to-one with SpeechXAI’s returned features and scores.
- The notebook rejects an utterance when its first aligned word begins before SpeechXAI’s 100 ms pre-word padding. The upstream removal function does not clamp this slice at zero, so accepting such an example could make the displayed mask differ from the perturbation actually evaluated.
- The final evaluation uses the top- k SpeechXAI words to define the shared masking duration.
- Overlapping padded masks are counted by their union duration so the audio-removal budget is not inflated.
- The notebook links the upstream implementation modules beside the attribution code. The LOO explainer calls single-word removal, the removal utility performs waveform replacement, and the faithfulness evaluator invokes multi-word removal. The linked files are `loo_speech_explainer.py`, `utils_removal.py`, and `faithfulness_measures_speech.py`.
- Notebook 01 writes no result files; all results remain visible in the executed notebook.

What was tested

The selected audio was loaded and classified, then transcribed and aligned with SpeechXAI’s WhisperX function. The returned segments were passed directly into SpeechXAI’s leave-one-out explainer and checked against the returned feature order. Timestamp validity, word-score alignment, and the 100 ms first-word safety condition were validated. The original prediction obtained from SpeechXAI’s pydub representation matched the prediction obtained from the waveform shown in the notebook. The first visualization used real word locations, effective silence masks, raw confidence-drop attribution values, and per-word transcription confidence. The second visualization inspected padded removal intervals and computed overlap depth at waveform-sample resolution. Top- k SpeechXAI deletion was compared with repeated random masking using the same union duration.

Outcome

The notebook executed successfully for utterance `Ses01F_impro01_F005`. WhisperX located the first word at 0.594 seconds, preserving the initial silence and later pauses. The returned word-confidence values ranged from 0.760 to 0.980. The classifier predicted anger with confidence 0.8781. SpeechXAI assigned the largest raw confidence drop, 0.4218, to “what’s”. The overlap diagnostic found six distinct regions covered by multiple padded masks, with a maximum depth of two simultaneous masks. The visualizations, attribution table, consistency checks, and shared-duration evaluation all completed successfully.

Remaining issues

The current validation covers one utterance only. WhisperX confidence is useful diagnostic information but is not a guarantee that every boundary is correct. The 100/40 ms expansion is present in the released Python code but is not documented in the paper, so future experiments should state explicitly whether they reproduce the repository behavior or use unpadded paper-level word intervals. Utterances with a word beginning inside the first 100 ms are currently rejected because of the upstream SpeechXAI slicing behavior; supporting them safely would require clamping the library’s perturbation boundary. The pipeline must be run on a controlled multi-audio set before drawing comparative conclusions.

3.2 Notebook 02 — Attention Access Test

Purpose

Determine whether the wav2vec 2.0 Transformer attention maps required by the LeGrad-inspired method are accessible.

Technical idea

The active classifier is a Hugging Face `Wav2Vec2ForSequenceClassification` model. The diagnostic step requests attention outputs directly from its Transformer encoder and inspects their dimensions and numerical behavior.

Decisions made

- Attention access is treated as a required checkpoint before developing LeGrad.
- The Hugging Face classifier is accessed directly, matching the model-loading path used by SpeechXAI.
- Attention tensors should be inspected across layers and heads and visualized for at least one layer-head pair.

What was tested

The configured SUPERB classifier was run on the same IEMOCAP utterance used in Notebook 01 with attention output enabled. The notebook inspected every returned layer, validated finite values, printed tensor dimensions, and visualized one final-layer attention head.

Outcome

Attention access succeeded. The model returned 12 attention tensors, one per Transformer layer, each shaped [1, 12, 201, 201] for the selected utterance.

Remaining issues

Notebook 03 must confirm that these returned attention tensors retain usable, nonzero gradients for the original predicted-class logit.

3.3 Notebook 03 — Attention Gradient Test

Purpose

Verify that gradients of a target class score with respect to Transformer attention maps can be obtained.

Technical idea

LeGrad uses sensitivity to attention maps as its explanation signal. For speech, the target quantity is the classifier score for the original predicted emotion, and the required diagnostic is $\partial s_{\hat{y}}/\partial A_l$ for one or more Transformer layers.

Decisions made

- The target is the original predicted-class logit or score.
- Returned attention tensors will retain gradients during backpropagation.
- Gradient shape, magnitude, finite values, and nonzero proportion will be checked before attribution aggregation.
- Logits are preferred over probabilities when probability saturation weakens gradients.

What was tested

The notebook scaffold and reusable gradient-retention utilities are present.

Outcome

The conceptual gradient path is defined. A real model forward and backward pass remains to be validated after Notebook 02 succeeds.

Remaining issues

Gradient availability depends on how the wav2vec 2.0 attention implementation returns intermediate tensors. Hooks may be required if returned attentions are detached from the computational graph.

3.4 Notebook 04 — LeGrad-Inspired Temporal Explanation

Purpose

Convert attention-gradient information into a temporal relevance signal over the speech utterance.

Technical idea

For each Transformer layer, positive class-sensitive attention gradients are combined with attention activations, averaged across heads, aggregated across layers, and reduced from a token-token matrix to token relevance. Token relevance is then mapped to audio time.

Decisions made

- The method is presented as LeGrad-inspired because it adapts the central attention-gradient principle from vision to speech.
- Relevance uses positive gradients multiplied by attention values.
- Head and layer aggregation use simple means for interpretability and reproducibility.
- Temporal outputs use small fixed-duration bins or token intervals that can participate in duration-matched deletion evaluation.

What was tested

Reusable aggregation and token-to-time utilities have been prepared.

Outcome

The representation and output format are defined. Real relevance curves depend on successful attention-gradient extraction in Notebook 03.

Remaining issues

The reduction direction of the token-token relevance matrix must be selected empirically and kept fixed. Temporal mapping must also account for the wav2vec 2.0 feature encoder stride.

3.5 Notebook 05 — Quantitative Evaluation

Purpose

Run the complete multi-audio comparison in one coherent experiment. This notebook owns batch explanation generation, common interval representation, equal-duration masking, deletion metrics, and aggregate quantitative plots.

Technical idea

The validated SpeechXAI workflow from Notebook 01 and LeGrad-inspired workflow from Notebook 04 are applied to the same controlled IEMOCAP set. SpeechXAI top- k words define a union duration X . SpeechXAI masks, LeGrad top bins, and repeated random bins are then compared after silencing exactly X seconds while preserving waveform length.

Decisions made

- Batch SpeechXAI generation is part of evaluation rather than a separate notebook because its single-audio behavior is already established in Notebook 01.
- Interval unification and masking are part of the evaluation operation rather than independent experimental outcomes.
- Both explanation methods use the common columns `audio_id`, `start`, `end`, and `score`.
- SpeechXAI is the source of duration X because its native unit is a variable-duration word.
- Top- k values are 1, 2, 3, and 5.
- LeGrad bins are selected by score until their union duration reaches X .
- Random bins use the same duration and are repeated 20 times.
- The main metric is absolute confidence drop for the original predicted class.
- Relative confidence drop and prediction flip rate are secondary metrics.
- Aggregate tables, deletion curves, boxplots, duration checks, and flip rates remain in this notebook because they are direct outputs of the quantitative experiment.

What was tested

Notebook 01 validated real SpeechXAI attribution, silence masking, union-duration accounting, confidence drop, and repeated same-duration random masking on one utterance. The consolidated Notebook 05 currently contains the protocol scaffold that will extend those validated operations to SpeechXAI and LeGrad across the evaluation set.

Outcome

The evaluation schema and fairness constraints are defined and validated at single-audio scale. The previous standalone SpeechXAI, interval-masking, deletion, and final-summary placeholders have been removed because their responsibilities now belong to this one quantitative experiment.

Remaining issues

The complete run requires a working LeGrad example from Notebook 04, a documented evaluation set, and execution across all selected audios.

3.6 Notebook 06 — Qualitative Examples

Purpose

Create interpretable examples that show where SpeechXAI and LeGrad agree or disagree and how masking changes model confidence.

Technical idea

Each example combines the waveform, word-level SpeechXAI intervals, the LeGrad temporal relevance curve, selected masking regions, and before/after classifier confidence.

Decisions made

- Examples will be selected from quantitative outcomes rather than chosen only for visual appeal.
- The final set should include agreement, LeGrad-favored, SpeechXAI-favored, and difficult cases.
- Interpretations will distinguish model faithfulness from human plausibility.

What was tested

The notebook structure and plotting requirements have been defined.

Outcome

The qualitative analysis plan is ready.

Remaining issues

Representative cases can be selected after Notebook 05 produces the complete deletion results. Aggregate tables remain in Notebook 05; this notebook is reserved for case-level interpretation.

4 Evaluation Design

The final quantitative evaluation follows these steps:

1. SpeechXAI returns importance scores for word-aligned audio segments.
2. For each audio and each $k \in \{1, 2, 3, 5\}$, the top- k SpeechXAI words are selected.
3. The selected words define a total duration X seconds.
4. Those SpeechXAI word intervals are replaced by silence, and the confidence drop for the original predicted class is measured.
5. LeGrad time bins are ranked by relevance, and the highest-ranked bins whose total duration equals X are selected.
6. Those LeGrad regions are replaced by silence, and the same confidence drop is measured.
7. Random time bins with total duration X are selected repeatedly to form a stable random baseline.
8. All methods use the same audio, classifier, target class, masking operation, and removed duration.

SpeechXAI naturally explains words, while the LeGrad-inspired method explains Transformer time bins or tokens. Equalizing the total occluded duration makes the comparison focus on where each method assigns relevance.

The main metric is:

$$\text{drop} = p_{\text{original}}(\hat{y}) - p_{\text{masked}}(\hat{y}),$$

where $\hat{y}$ is the class predicted from the original audio. Secondary metrics are relative confidence drop and prediction flip rate. Mean masked duration is reported for every top- k setting.

5 Methodological Decisions Summary

Component	Decision	Reason
Dataset	IEMOCAP speech, transcripts, and emotion labels	It is a standard speech emotion resource and provides the audio and annotations required by the selected classifier.
Classifier	SUPERB wav2vec2-base-superb-er checkpoint used by SpeechXAI	Using the same classifier removes model choice as a confound when comparing SpeechXAI and LeGrad-inspired explanations.
Model label space	Anger, happiness/excitement, neutral, sadness	These are the four outputs supported by the selected pretrained classifier.
Target class	Original predicted class	Faithfulness is evaluated relative to the model decision being explained.
SpeechXAI unit	Word-aligned audio segments	This is SpeechXAI’s native, human-readable explanation unit.
LeGrad unit	Transformer tokens or fixed temporal bins	Attention-gradient relevance is naturally produced at the encoder-token level.
LeGrad formulation	Positive attention gradients combined with attention values, aggregated across heads and layers	This preserves the central LeGrad idea while adapting its output to speech time.
Shared duration	X seconds from SpeechXAI top- k words	This makes word-level and time-bin explanations comparable under equal audio removal.
Top- k values	1, 2, 3, and 5 words	These settings provide a compact deletion curve while respecting SpeechXAI’s native output.
Masking method	Silence replacement with unchanged waveform length	It provides a simple perturbation and avoids duration changes that could independently affect the classifier.
Random baseline	Same-duration random bins, repeated 20 times	Repetition estimates baseline variance and provides a stable reference.
Main metric	Confidence drop for the original predicted class	It measures how strongly the selected regions support the model’s original decision.
Secondary metrics	Relative confidence drop and prediction flip rate	They complement absolute confidence drop across samples with different confidence levels.
Notebook 01 scope	One utterance, in-memory outputs	It establishes an understandable baseline and avoids mixing pipeline validation with dataset-scale evaluation.